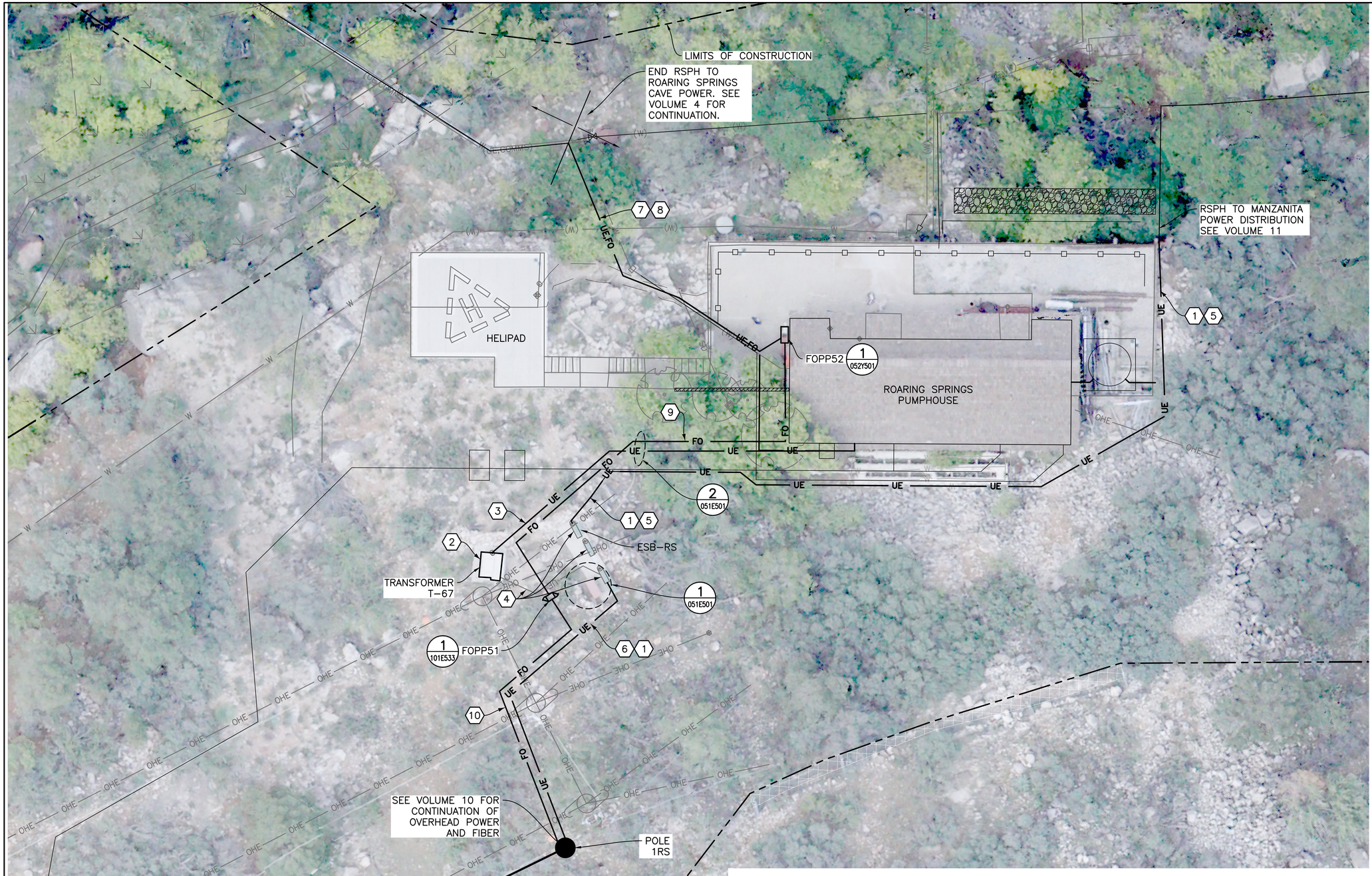

KEY NOTES: (X)

1. 3-PHASE MEDIUM VOLTAGE UNDERGROUND POWER CABLE REPLACEMENT. INSTALL TRENCH AND UNDERGROUND POWER CABLE FROM EXISTING ROARING SPRINGS ENCLOSURES. TERMINATE MEDIUM VOLTAGE UNDERGROUND CABLES WITH 200AMP LOADBREAK ELBOWS. UTILIZE EXIST JUNCTION POSITIONS IN EXIST ENCLOSURES.
2. REPLACE EXIST 500KVA TRANSFORMER WITH A 750KVA,12.47/7.2KV – 480Y/277V TRANSFORMER (T-67). RE-ATTACH EXIST MV CABLE TERMINATIONS. INSTALL 15KV CLASS, 8.4KV MCOV LOADBREAK ELBOW SURGE ARRESTERS ON OPEN TAP POSITIONS AND LEAVE SWITCH CLOSED FOR BOTH POSITIONS. REFER TO SHEET 008E509 FOR TRANSFORMER DETAILS. INSTALL GROUND RING AT TRANSFORMER WITH TWO GROUND RODS AND TWO BONDS TO TRANSFORMER. MATCH MATERIALS USED FOR GROUNDING ELECTRODE AT PUMP STATION BUILDING.
3. INSTALL (3) SECONDARY POWER CIRCUITS FROM TRANSFORMER TO RSPH SERVICE ENTRANCE. REFER TO ONELINE DIAGRAM ON SHEET 052E601 FOR CABLE AND CONDUIT SIZE. SEE SHEET 051E501 FOR SECONDARY TRENCH DETAIL.
4. EXISTING MEDIUM VOLTAGE POWER CONDUCTORS BETWEEN EXIST ENCLOSURES AND TRANSFORMER TO REMAIN IN PLACE. RE-TERMINATE CONDUCTORS AT TRANSFORMER.
5. (1) 15KV (3/C) 1/0 AWG ALUMINUM ARMORED POWER CABLE.
6. (3) 15KV (1/C) 4/0 AWG URD CABLE WITH CONCENTRIC NEUTRAL IN (3) 3" CONDUIT (1 PHASE/CABLE PER CONDUIT). SEE SHEET 08E507 FOR JOINT USE FIBER AND MV POWER TRENCH DETAIL.
7. POWER TO RS CAVE MINI POWER ZONE. (2) #4, #4 GND, 2"C.
8. ½" ARMORED 24-STRAND SINGLEMODE FIBER CABLE IN 2" CONDUIT UNDERGROUND.
9. INSTALL 144-STRAND SM FIBER IN ORANGE TUBE OF NEW 7-WAY MICRODUCT.
10. INSTALL 144-STRAND SM FIBER IN 2" CONDUIT UNDERGROUND.

10 0 10 20 1" = 10'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. VANG CADD R.D. CLARK TECH. REVIEW: P. DAVILA DATE: JUNE 2026	SUB SHEET NO. 051 E101	TITLE OF SHEET SITEWORK ELECTRICAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 30 OF 113

10/13/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\103505445_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\051-E101.dwg